

# Evaluating Human Impressions of an Initiative-taking Robot

Priscilla Kan John\*  
priscilla.kanjohn@anu.edu.au  
The Australian National University  
Canberra, Australia

Wan Zhu  
wan.zhu@anu.edu.au  
The Australian National University  
Canberra, Australia

Xuanying Zhu\*  
xuanying.zhu@anu.edu.au  
The Australian National University  
Canberra, Australia

Tom Gedeon  
tom.gedeon@curtin.edu.au  
Curtin University  
Perth, Australia

## ABSTRACT

Social robots can be applied in different settings to enhance experiences for humans. Understanding physical and behavioural aspects of robot design is important to promote acceptance in human-robot interaction. In this study, we focus on robot initiative-taking as a behavioural aspect and investigate how it influences human perception and emotion during human-robot interaction. We built on previous work using questionnaires to evaluate user impressions in the presence or absence of robot activeness (initiative-taking). We also used Galvanic skin response (GSR) as a physiological measure to gauge participants' emotion during interaction. Questionnaire analysis confirmed that generally active robot behaviour improved human impression. Moreover, the order of interaction seems to matter for participants to take more notice of robot initiative-taking. GSR analysis supported the questionnaire results, showing that on average participants were more emotionally aroused during an active interaction and that order of interaction somewhat mattered.

## CCS CONCEPTS

• **Human-centered computing** → *Collaborative and social computing*; • **Computer systems organization** → *Robotics*.

## KEYWORDS

Human robot interaction, Impression evaluation, Physiological response, Autonomy

### ACM Reference Format:

Priscilla Kan John, Xuanying Zhu, Wan Zhu, and Tom Gedeon. 2022. Evaluating Human Impressions of an Initiative-taking Robot. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29-May 5, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3491101.3519633>

\*Both authors contributed equally to this research.

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CHI '22 Extended Abstracts, April 29-May 5, 2022, New Orleans, LA, USA

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ACM ISBN 978-1-4503-9156-6/22/04...\$15.00

<https://doi.org/10.1145/3491101.3519633>

## 1 INTRODUCTION

With advances in technology, humans interact more and more with machines to accomplish various tasks. Robots as machines “capable of carrying out a complex series of actions automatically” [9] are becoming more prevalent in everyday life. The field of Human Robot Interaction (HRI) is concerned with understanding, designing and evaluating robotic systems for use by or with humans [15]. In order to improve user experience and design of robots, studies are needed on evaluating the impressions a robot leaves on people. These will help with designing robot behaviour that is preferable to humans and therefore result in a better interaction experience. We are especially interested in human-robot interaction in a social context. Given the increasing uptake of social robots [22], there is a greater need to understand the social aspects of robot behaviour. This paper aims to investigate users' emotions and impressions when a robot takes initiative (i.e. displays “active” behaviour). We collected experimental data on participants' impressions (through questionnaires) and emotions (through physiological measure) as they interact with a Rapiro humanoid robot [19] in two situations, one where the robot initiated a greeting and one where it did not. We analysed collected data to evaluate the impression left on participants and draw insights for robot design.

## 2 MOTIVATION AND BACKGROUND

### 2.1 Human Robot Interaction

Humans have been fascinated by robots for centuries, with various references of robot-like behaviour and implications for humans appearing in religion, mythology, philosophy and fiction [15]. Nowadays, machines in various forms help us in our everyday life and we expect to see this trend continue, bringing us more sophisticated robots to help in areas such as aged care, education, or home assistance [17, 28, 33]. This trend has led to the emergence of human-robot interaction (HRI) as a multi-disciplinary research area at the intersection of fields such as robotics, cognitive science, and human-computer interaction [15, 33].

### 2.2 Robot Behaviour

Using robots in a social context, e.g. in aged care, education, or domestic settings, can be greatly beneficial to humans. However, how these robots should behave and interact with humans is still largely not well understood [3]. Bartneck and Forlizzi [3] define a social robot as “an autonomous or semi-autonomous robot that interacts and communicates with humans by following the behavioural norms

expected by the people with whom the robot is intended to interact” [3]. Behavioural norms can vary between different cultures and will therefore change with social context. Humans generally tend to expect robots to have human-like social characteristics during interaction [13, 26]. One aspect of social interaction that is of interest is the role the robot takes as an active or a passive participant during the interaction and how it is perceived by humans in those situations. Taking initiative is often seen as an aspect of autonomy in robots [3].

### 2.3 Emotion Measures in HRI

The role of emotions is significant in human behaviour, interaction and communication [13]. They are an important consideration in HRI because of the recognition that people tend to treat robots as they tend to treat other people [13]. To evaluate the effect of HRI on users’ emotions, we need appropriate tools to measure emotions. Subjectively, emotions can be measured by using psychological questionnaires. Emotions can also be measured objectively by monitoring physiological and physical changes. Questionnaires are quick and convenient and do not require additional devices. However, they may not always be reliable as true emotions may not always be expressed. Hence, using physiological measures helps uncover at least the magnitude of true emotional reactions. For example, skin conductance, also known as electro-dermal response, measures the electrical conductance of an individual’s skin, which varies with the amount of sweat on the skin. Skin conductance contains two types of activity, one being tonic and the other being phasic [6]. The tonic component refers to continual and slow-moving changes caused by body temperature or external temperature. The phasic components are quicker pulse-like activities, often seen as peaks in recorded data. The amplitude of these peaks is linearly associated with emotional arousal, the intensity of the emotion [5, 6, 20]. Skin conductance is not under conscious control but is instead autonomously modulated by sympathetic activity which drives human behaviour, cognitive and emotional states on a sub-conscious level. Therefore, skin conductance offers direct insights into autonomous emotional regulation [11]. The fluctuations of skin conductance can be measured via Galvanic skin response (GSR) by calculating electrical potential between electrodes placed on surfaces of the skin.

### 3 RELATED WORK

Humans tend to apply social rules in their interaction with robots despite realising that the latter are machines [26], making it interesting to study how different robot behaviours impact human impressions. *Active behaviour* is defined as “*behaviour in which a robot takes the initiative and does something towards a human user without any prior commands or instructions from the user*” [1]. Conversely, the behaviour in which a robot reacts to user’s actions is defined as *passive behaviour*. Nakata et al. [25] designed a robot to react after people touched the robot with their hands. They evaluated the impressions that people had after seeing the reaction of the robot. The results showed that people had a positive impression of the robot when the robot interacted with them in a way that professed a kind of emotional attachment to them [25] as cited in [1]. Kakio et al. [18] designed different robot behaviours

as possible reactions when the robot was pushed and investigated the differences in users’ impressions depending on these reactions. Both of these studies on user’s impressions focused on the passive behaviour of the robots.

Akiho and Sugaya [1] investigated users’ impressions of the robot when it showed active behaviour. In their experiment, they evaluated participants’ impressions in two different situations: one where the robot took initiative to greet them and the other where it did not initiate the greeting. Participants were divided into 2 groups, each with 5 participants, where both groups experienced one situation with active behaviour, and one without but in a different order. They examined the differences in users’ impressions depending on whether active behaviour was displayed or not by the robot. They found that user impressions were significantly more positive when the robot took initiative and users felt greater affinity with the robot and had a greater desire to continue the interaction. However, the sample size (10 participants) was relatively small and the questionnaire used did not contain standardised metrics for evaluating human impression.

Baraglia et al. [2] tried to figure out when a robot should take initiative for efficient human-robot collaboration in a joint task execution. They tested three main scenarios: (1) humans could initiate getting help from the robot, (2) the robot initiated help reactively when it detected that help was needed, (3) the robot initiated help proactively whenever it could help. The study found that people collaborated best with a proactive robot but preferred to have control of when to get help.

There are overall less studies on the evaluation of human impressions when a robot takes the initiative. This paper seeks to add to the literature in this area, building on the work by [1].

## 4 APPROACH

Our experimental set-up is based on and extends what is done in [1]. In [1], users’ impression of a robot is investigated in the presence or absence of active behaviour in a greeting scenario. We kept the principles of the active and passive interaction in [1] but implemented them differently. For the passive greeting, instead of the robot reacting to being touched, we used a voice interaction where the robot responds to a greeting by the participant. For the active greeting, the robot initiated the greeting conversation. Instead of using sensors to detect human actions, we used a Wizard of Oz (WOz) method to trigger the robot’s actions at various stages. WOz is a technique where participants are given the impression of autonomous functionalities in a robot but these functionalities are not fully developed yet and are actually implemented by a human. WOz is often used in HRI to allow for a faster and lower-cost experimental set-up (see examples in [8, 16, 23]). As robotic system, we use a Rapiro robot [19] with a mounted raspberry pi board [14]. Our experiment differs from [1] in that we recruited a larger number of participants, included the Godspeed questionnaire and analysed GSR as a proxy for emotional arousal.

### 4.1 Study Design

We wanted to determine if there is a difference in the impression and emotion experienced by humans when a robot took initiative (showed ‘activeness’) compared to when it did not. If there is a

difference, we wanted to explore how it played out across a number of characteristics associated with the robot. The human-robot interaction consisted of a greeting component first then a dancing component before ending in farewell. Two experimental scenarios were explored:

- (1) The robot initiated the greeting and the participant responded.
- (2) The robot did not initiate the greeting, but the participant did, and the robot responded.

In both situations the robot then invited the participant to dance with it after the greeting component and ended with saying goodbye.

The experiment consisted of two groups of participants (A and B) and followed a crossover design format [35]. This allowed us to test for treatment as well as order effects with limited experimental resources while reducing error variance. Each group experienced both experimental scenarios described above but in different order. Group A experienced scenario (1) first, followed by scenario (2). Group B experienced scenario (2) first, followed by scenario (1).

Akiho et al [1] only included a greeting function and to mitigate against the interaction being too short, they repeated their experiment three times for each participant to give them a chance to form a complete and concrete impression of the robot. Instead, we chose to lengthen the interaction with a dancing component, a natural next step in the hierarchy proposed by [21] as reported in [1].

Sixteen participants took part in the experiment under a study with human ethics approval at the Australian National University. The participants were recruited by inviting friends of one of the authors (convenience sampling). The participant group was made up of 7 females (43.8%) and 9 males (56.2%) aged from 19 to 23 (mean = 21.4, standard deviation = 1.0). Participants were randomly assigned to either Group A or B so each group consisted of 8 participants. On average, participants spent 95.3 ( $\pm$  19.4) seconds on scenario (1) and 81.8 ( $\pm$  11.2) seconds on scenario (2).

## 4.2 Experimental Procedure

In the experiment, the following sequence of events were used:

- (1) The participant walked towards the robot.
- (2) The robot turned its head around to look at the participant.
- (3) One of the experimental greeting scenarios was explored:
  - (i) scenario 1: the robot greeted the participant first, or
  - (ii) scenario 2: the participant was prompted to greet the robot first and the robot responded.
- (4) The robot then performed a dance and invited the participant to join.
- (5) The robot bid farewell to the participant.
- (6) The participant walked away from the robot.
- (7) The participant was asked to fill in a questionnaire to evaluate their impression of the robot.

## 4.3 Research question and Hypotheses

We explored the following research questions: (1) Does an initiative-taking robot produce a better impression on humans than one that does not take the initiative? if yes, (1.1) How different? (2) Does it make a difference if humans experience an initiative-taking robot first, then experience the same robot without the initiative-taking? (3) Do participants experience a higher state of emotional

arousal (a state of heightened electrodermal activity) when they are interacting with an active robot compared to a more passive robot.

These led to the following hypotheses:

- H1.** Activeness of robot will produce a better impression on humans.
  - 1.1 Participants will have a more positive impression of an active robot as measured by a number of constructs in a questionnaire.
- H2.** The order of interaction matters.
- H3.** Physiological measures such as GSR will support hypotheses 1 and 2.
  - 3.1 Participants' GSR when the robot takes the initiative show higher arousal than when the robot does not.
  - 3.2 Participants' GSR are different when orders are different.

## 4.4 Data Collection

We used questionnaires to collect data on participants' impressions during their interaction with the robot. Questions from [1] with a four-point likert scale, relating to activeness (if participants thought the robot was active), pleasantness (if participants thought the robot was trying to entertain them), affinity (if participants had a pleasant feeling with the robot), intentionality (if participants thought the robot had some sort of intention) and continuity (if participants wanted to continue to play with the robot), were used. An aggregated overall impression measure was obtained by summing the aforementioned five measures. Questions from the Godspeed questionnaire [4] were also added, which evaluate the following constructs using a five-point semantic differential scale: anthropomorphism, animacy, likeability, perceived intelligence and perceived safety.

We also measured Galvanic skin response (GSR) using Empatica E4 wristband [10] during interaction with the robot.

## 5 ANALYSIS AND RESULTS

### 5.1 Analysis of responses from questionnaires

Paired tests were done to investigate within-group differences for measures in both sets of questionnaires. For each group of participants, responses after interaction 1 was compared with responses after interaction 2 for each measure. The distribution of the responses was tested for normality first. If both distributions being compared are normal, then paired t-test is used. If even one distribution being compared is not normally distributed, the Wilcoxon signed rank test is used. Two-sample difference test were done to investigate if order of interaction mattered: the active interactions in Group A and Group B were compared and likewise the passive interactions in Group A and Group B. The distribution of the responses was tested for normality first. If both distributions being compared are normal, then a two-sample t-test (Welch test) is used. If even one distribution being compared is not normally distributed, the Mann-Whitney rank test is used. We also ran paired difference tests between first and second interactions for all participants, as well as between active and passive interactions for all participants, to check if there was any significant differences between them.

**5.1.1 Akiho et al [1] questionnaire analysis:** As indicated in Table 1, for Group A paired test comparison, statistical significance

**Table 1: Results of Statistical Analyses on Questionnaires**

| Questionnaire    | Measure                | Group A<br>Paired Test | Group B<br>Paired Test | Independent Group Test<br>Comparing Group A & B | Interaction Order<br>Comparison | Active and Passive<br>Comparison |
|------------------|------------------------|------------------------|------------------------|-------------------------------------------------|---------------------------------|----------------------------------|
| Akiho et al. [1] | Activeness             | p<0.01                 | p<0.05                 | -                                               | -                               | -                                |
|                  | Pleasantness           | -                      | -                      | -                                               | -                               | -                                |
|                  | Affinity               | -                      | -                      | -                                               | -                               | -                                |
|                  | Intentionality         | -                      | p<0.05                 | -                                               | -                               | -                                |
|                  | Continuity             | -                      | p<0.01                 | -                                               | -                               | -                                |
|                  | Overall Impression     | p<0.01                 | p<0.01                 | -                                               | -                               | p<0.01                           |
| Godspeed [4]     | Anthropomorphism       | -                      | p<0.01                 | -                                               | p<0.05                          | -                                |
|                  | Animacy                | -                      | p<0.05                 | -                                               | -                               | p<0.01                           |
|                  | Likeability            | p<0.05                 | -                      | -                                               | -                               | p<0.01                           |
|                  | Perceived Intelligence | -                      | -                      | -                                               | -                               | -                                |
|                  | Perceived Safety       | -                      | p<0.05                 | -                                               | -                               | p<0.05                           |

-: Not statistically significant

was found in activeness (p<0.01 using Wilcoxon sign rank, signed rank=28) and for overall impression (p<0.01 using t-test). No statistical significance was found for pleasantness (p=0.375), affinity (p=0.062), intentionality (p=0.375) and continuity (p=0.125), all using Wilcoxon signed rank test.

For Group B paired test comparison, statistical significance was found for activeness (p<0.05 using Wilcoxon sign rank, signed rank=0), intentionality (p<0.05 using Wilcoxon signed rank, signed rank=0), continuity (p<0.05 using t-test, t=-2.65, df=7) and for overall impression (p<0.01 using t-test, t-stat=-5.98, df=7). No statistical significance was found for pleasantness (p=0.125) and affinity (p=0.125), both using Wilcoxon sign rank.

Independent group test for both active and passive sessions comparing Group A and B showed there were no statistically significant differences for questions from [1]. Comparison between first and second interactions for all participants showed no statistical significance for any of the measures.

Comparison between all active and all passive interactions, for all participants, irrespective of order showed no statistical significance for all the measures except for overall impression (p<0.01, t-stat=4.63, df=15).

**5.1.2 Godspeed Questionnaire analysis:** As indicated in Table 1, for Group A paired test comparison, statistical significance was found for likeability (p<0.05, using t-test, tstat=2.69, df=7). There was no statistical significance found for anthropomorphism (p=0.658), animacy (p=0.285), perceived intelligence (p=0.785) and perceived safety (p=0.197), all using t-test.

For Group B paired test comparison, statistical significance was found for anthropomorphism (p<0.01, using t-test, t-stat=-4.98, df=7), animacy (p<0.05, using Wilcoxon sign rank, signed rank=1) and perceived safety (p<0.05, using t-test, t-stat=-2.97, df=7). No statistical significance was found for likeability (p=0.129) and perceived intelligence (p=0.064).

Independent group test for both active and passive sessions comparing Group A and B showed there were no statistically significant differences for questions from the Godspeed questionnaire

[4]. Comparison between first and second interactions for all participants showed no statistical significance for the measures in the Godspeed questionnaire, except slightly for anthropomorphism (p=0.048, t-stat=-2.15, df=15).

Comparison between all active and all passive interactions, for all participants, irrespective of order showed statistical significance for animacy (p<0.01, t-stat=3.13, df=15), likeability (p<0.01, t-stat=3.18, df=15) and perceived safety (p<0.05, t-stat=2.45, df=15). No statistical significance was found for anthropomorphism (p=0.237) and perceived intelligence (p=0.941).

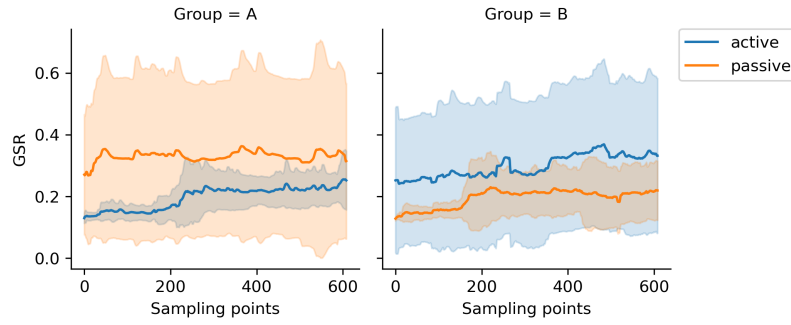
## 5.2 Analysis of GSR responses

**5.2.1 Timeline Analysis of GSR signal.** We plotted mean GSR to get an idea of how the average GSR signal fluctuates over time throughout an interaction, as Figure 1 shows. As the length of each interaction varied, we performed linear interpolation [34] on shorter GSR segments to match the length of the longest segment.

Figure 1 shows that the GSR signal for each group is distinct for the active interaction and for the passive interaction. We expected the GSR signal for the ‘active’ interaction to be higher on average to the signal for the ‘passive’ interaction. Instead, we see that the average signal is higher for the passive interaction in Group A where participants experienced active interaction first. Given that GSR can accumulate when two sessions occur consecutively [6], and group A participants experienced an active interaction followed by a passive interaction, the higher GSR level during the passive interaction may be caused by an accumulation from the active interaction which participants experienced first.

**5.2.2 Statistical Analysis of GSR features.** To verify the GSR accumulating phenomenon and to explore the differences of GSR features by activeness and by order, we conducted paired statistical tests on each GSR feature similarly as specified in section 5.1.

To extract features, we first preprocessed GSR data for noise removal, normalisation and segmentation. We segmented the GSR by each interaction session, so that each segment corresponds to one participant’s GSR state data elicited by one interaction with the robot. After preprocessing GSR, we extracted 14 features as



**Figure 1: Relative changes in normalised GSR signal averaged across all participants.**

listed in Table 2. The first 8 features were extracted following [30], which have been popularly used for emotion recognition. The last 6 features were extracted following [20], as they measure the phasic components of GSR, which are found to be associated with the perceived emotion arousal [5, 6].

For Group A paired test comparison, two features were statistically higher in active interactions than in passive interactions. These two features were the number of peaks in low-passed (LP) GSR ( $p < 0.05$  using t-test) and the number of peaks in very low-passed (VLP) GSR ( $p < 0.05$  using Wilcoxon sign rank, signed rank=25). No statistical significance was found for other features.

For Group B paired test comparison, statistical significance was not found for any of the examined features between active interactions and passive interactions.

Independent group test for both active and passive sessions comparing Group A and B showed that the number of peaks in VLP GSR and the ratio of peak occurrences in VLP GSR to peaks in LP GSR were statistically higher in passive sessions of Group B than in passive session of Group A ( $p < 0.05$ , t-test in both cases).

Comparison between first and second interactions for all participants showed no statistical significance for any of the features.

Comparison between all active and all passive interactions, for all participants, irrespective of order showed that two features were statistically higher in active interactions than in passive interactions. These two features were the number of peaks in LP GSR ( $p < 0.05$ , Wilcoxon, signed rank=25) and the number of peaks in VLP GSR ( $p < 0.01$ , Wilcoxon sign rank, signed rank=25). No statistical significance was found for other features.

**5.2.3 Classification between Active and Passive Interactions.** We built several classifiers to explore if we can differentiate active and passive interactions based on GSR signals. We extracted 14 features as listed in Table 2, based on the literature [20, 30].

To differentiate active and passive interactions using GSR features from participants, we examined three classifiers, including 1) Support Vector Machine (SVM), 2) Random Forest (RF) and 3) Neural Network (NN), due to their success in recognising emotion [29]. Each classifier was trained using 3-fold leave-n-subjects-out cross validation which has previously been used in [31, 32] for physiological signal classification. Thus, data from one third of participants were used as the testing set and data from the remaining two thirds of participants were used as training set.

**Table 2: Extracted features from GSR signal**

| Extracted Features                                           |
|--------------------------------------------------------------|
| 1. Minimum                                                   |
| 2. Maximum                                                   |
| 3. Mean                                                      |
| 4. Standard deviation                                        |
| 5. Variance                                                  |
| 6. Root mean square                                          |
| 7. Means of the first derivatives                            |
| 8. Means of the second derivatives                           |
| 9. Number of peaks in normalised GSR                         |
| 10. Number of peaks in low-passed (LP) GSR                   |
| 11. Number of peaks in very low-passed (VLP) GSR             |
| 12. Mean of amplitudes of peaks in LP GSR                    |
| 13. Mean of amplitudes of peaks in VLP GSR                   |
| 14. Ratio of peak occurrences in VLP to peaks in LP (VLP/LP) |

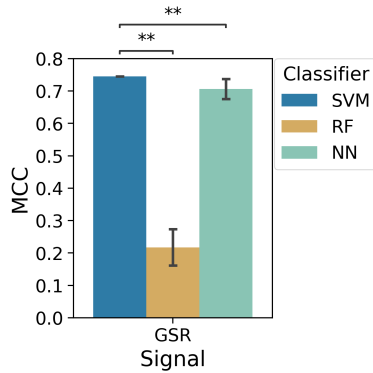
Following [12] we trained each model with 10 different runs and calculated the mean accuracy and Matthews Correlation Coefficient (MCC) [24] over the 10 runs on the test set. MCC returns a value between -1 and 1, where -1 shows a complete mismatch between predictions and observations, 0 denotes a random prediction, and 1 indicates a perfect prediction. It considers the balancing ratio of all elements in the confusion matrix and thus is a more reliable than precision, recall and F1 score [7].

The average results of the classification models on GSR over 10 runs are shown in Table 3. The best results were achieved by a SVM, with an accuracy of 85.7% and an MCC score of 0.75. The MCC score is significantly above the chance level, indicating a strong correlation between the extracted GSR features and the activeness of the interaction.

Following [27] we conducted the Wilcoxon signed-rank test between the MCC scores achieved by SVM and other models. The results are shown in Figure 2. The statistical analysis showed that the SVM significantly outperformed the RF and NN ( $p < 0.001$  in both cases).

**Table 3: Classification results from GSR**

| Classifier | Accuracy | MCC  |
|------------|----------|------|
| SVM        | 85.7%    | 0.75 |
| RF         | 60.7%    | 0.22 |
| NN         | 85.0%    | 0.71 |

**Figure 2: Mean MCC scores of each model when differentiating active and passive interactions.**

## 6 DISCUSSION

### 6.1 Hypothesis 1: Activeness of robot will produce better impression on humans

Overall, our results support hypothesis 1. Within-group test results presented in sections 5.1.1 and 5.1.2, both show that participants' total impression of the robot was significantly better when they experienced the interaction with active greeting. Moreover, in section 5.1.2 we found that participants tended to find the robot more likeable in the active interaction compared to the passive interaction when they experienced the active interaction first.

When participants experienced the passive interaction first, followed by active, they tended to find the robot more human-like and animated (weak significance), and felt safer (weak significance) with the robot in the active interaction. Participants also had a stronger perception that the robot had some intent (intentionality) and more desire to continue interacting with the robot (continuity) when experiencing active interaction after the passive one. These differences were not significant when participants experienced active followed by passive behaviour.

### 6.2 Hypothesis 2: The order of interaction matters

The independent group testing shows that participants had similar impressions in the active interaction they experienced irrespective of the order in which they experienced it. Likewise, participants had similar impressions in the passive interaction they experienced, irrespective of order. This indicates that order of interaction does not matter to the impression that participants had of the robot within either the active or passive interaction.

However, within-group testing showed that experiencing passive interaction first, as discussed in section 5.1, tended to make participants notice the active behaviour more, highlighting traits like intentionality, anthropomorphism, animacy and perceived safety, and making it more desirable to continue interacting with the robot.

### 6.3 Hypothesis 3: Physiological measures such as GSR will support hypotheses 1 and 2

Statistical analysis of GSR features indicate that some of the features that measure the phasic components of GSR (which are associated with perceived emotional arousal) support both hypothesis 1 and hypothesis 2. Notably, the analysis suggested that activeness of the robot produced a better impression on humans and that the order of interaction seemed to matter. Moreover, the classification model applied to GSR signals showed that active and passive interactions can be differentiated. While time analysis of GSR signals showed that participants were more emotionally provoked during the second interaction they experienced irrespective of whether it was active or passive, it is likely that it was caused by an accumulation of GSR during the second interaction, for which we did not account for in experimental design but can learn from.

### 6.4 Limitations

The sample size for our study may be too small and may not be sufficiently representative of the underlying population characteristics. Participants were mostly from the same Asian cultural background and had a similar technical background in computer science. Therefore results obtained in this study offer interesting insights but cannot be extrapolated to the general population with different cultural and technical backgrounds. Another limitation was that the rest time between the two interaction sessions for each participant may not have been long enough, leading to accumulation of GSR signal for participants and causing bias in the results of the time analysis.

## 7 CONCLUSION

We presented findings from our study in which we explored how initiative-taking by a robot affected human emotion and perception of the robot in a greeting and entertainment context during human-robot interaction. Participants reported an overall better impression of the robot when it took initiative. Participants each experienced both a scenario with active greeting behaviour and one without, but the order of the interaction was varied. Results show that the order of interaction could matter and suggest that if we want users to take more notice that the robot is taking initiative, we may want to design its behaviour so that it shows passive behaviour first before active behaviour. GSR data indicated that participants experienced stronger emotional arousal in the active interaction with the robot compared to the passive one. Further investigation is needed to explore the link between reported participant impression and measured emotional response, including to what extent order of interaction affected participants.

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